

Two Approximations Under One Name: Separating Sobolev Line Transfer from Expansion Opacity in Kilonova Ejecta

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ABSTRACT

Kilonova radiative-transfer codes routinely describe their line treatment as “the Sobolev approximation”, usually meaning its binned expansion-opacity form. We show these are two approximations with very different error budgets, and measure each separately. The harness is deliberately restricted—a 1-D homologous shell, pure absorption with fixed LTE populations, shell velocities of 1000–3000 km s^{−1}—because that restriction is what makes the two errors separable. Three independent calculations of one model—analytic Sobolev theory, a deterministic frequency-resolved solver, and the Monte Carlo code SEDONA—agree to $\sim 1\%$ on multi-line problems once a thermal-emission convention is matched. On a forest of 153 calibrated La II lines from the GSI database, the *expansion-opacity implementation* carries a strength-set error floor from its per-resonance substitution $\tau \rightarrow 1 - e^{-\tau}$, growing from $\sim 3\%$ for weak-line forests to +38–48% for strong ones and surviving undiminished to a Doppler width of 1 km s^{−1}, so no refinement of line widths rehabilitates it. It is invariant under transport bin width from 0.025 to 41 lines per bin, and so is intrinsic to the formalism rather than an artifact of use outside its design regime. The *Sobolev approximation itself* shows no strength floor and is accurate to $\lesssim 2\%$; its only measured failure is geometric, where a resonance falls near a boundary of the finite line-forming region. Across 36 conditions spanning four wavelength windows, three epochs and three ion mixtures the separation is universal, controlled by the realized maximum line optical depth at fixed geometry and line width. The shortcut in common use is thus wrong by tens of percent, in pure absorption, for reasons largely unrelated to the approximation it is named after.

Key words: radiative transfer – opacity – methods: numerical – stars: neutron – nuclear reactions, nucleosynthesis, abundances

1 INTRODUCTION

1.1 Kilonovae in three paragraphs

When two neutron stars spiral into each other and merge, a small fraction of their matter—between roughly a thousandth and a few hundredths of a solar mass—is flung outward at speeds of ten to thirty percent of the speed of light. This ejected matter is one of the very few environments in the universe dense enough in free neutrons to build the heaviest elements (gold, platinum, uranium, and the lanthanide row of the periodic table) through the rapid neutron-capture process, or *r-process*. The radioactive decay of these freshly made nuclei heats the expanding cloud, which then glows for days to weeks: this glow is a *kilonova* (for a review see Metzger 2020). One has been observed in detail, AT2017gfo, the optical counterpart of the gravitational-wave event GW170817 (Abbott et al. 2017), whose modelling gave the first direct evidence that such mergers synthesize heavy r-process elements (Kasen et al. 2017).

Everything we want to learn from a kilonova—how much matter was ejected, how fast, and above all *which elements*

were made—must be read out of its light. The light escapes only after filtering through the ejecta, and the filtering is dominated by an enormous number of atomic absorption lines. Lanthanide ions are the worst offenders: because of their complex open 4f electron shells, a single lanthanide ion can have tens of thousands of energy levels and millions of radiative transitions (Tanaka et al. 2020; Flörs et al. 2026b). Their presence raises the ejecta opacity by orders of magnitude and reddens and lengthens the transient (Barnes & Kasen 2013). The opacity of the ejecta, and therefore the color, brightness, and spectral features of the kilonova, is controlled by these line forests.

Simulating light transport through millions of lines, each intrinsically narrower than one part per million in frequency, is prohibitively expensive if done directly. Practical kilonova radiative-transfer codes therefore lean on an approximation introduced by Sobolev (1960) for expanding stellar envelopes. It reaches them in two distinct packagings. One family applies it line by line inside a Monte Carlo scheme with macroatom or downbranching redistribution (Lucy 2002, 2003; Kerzendorf & Sim 2014); the other bins it into the “expansion opacity” of Karp et al. (1977) and Eastman & Pinto (1993), which is what we test here. Section 2 explains both ideas from scratch. The distinction matters for how our results should be read, and we

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return to it in §6. The approximation is standard, venerable, and almost never checked against a resolved calculation *under kilonova conditions*. This work begins that check.

It also draws a line the field has left blurred. Sobolev’s approximation proper attenuates each line by $e^{-\tau s}$ and rests on two assumptions: that the gas does not change across a resonance, and that lines do not overlap. Expansion opacity is a further, and separate, step—a statistical device that bins many lines into an effective opacity and, as we show, substitutes $1 - e^{-\tau}$ for τ at every resonance crossing. Codes and papers commonly report using “the Sobolev approximation” when they mean the second. Our central result is that the two carry error budgets differing by roughly an order of magnitude, so the distinction is not pedantry.

1.2 What this paper does

We do not build a kilonova pipeline. We build the smallest apparatus that can measure the two approximation errors cleanly and *separately*, and we operate it on progressively more realistic inputs. Concretely, this first paper reports:

- a **three-way validation harness** (Section 3): the same spherically expanding model, atomic data, and level populations evaluated by (a) analytic Sobolev theory, (b) a purpose-written deterministic frequency-resolved solver, and (c) the Monte Carlo code SEDONA (Kasen et al. 2006) in both its resolved and its expansion-opacity modes;

- **verification experiments** (Section 4) in which the resolved-physics calculations agree at the $\sim 1\%$ level on the multi-line problems, and at the few-percent level on a single isolated line, establishing the baseline everything else is measured against;

- **the separation itself** (Section 5.3), the central result: on a forest of 153 calibrated La II lines from the GSI database (Flörs et al. 2026b,a), the Sobolev approximation errs by $+2.1\%$ in band flux while expansion opacity errs by $+44.8\%$, and the two fail through different mechanisms;

- **validity maps** (Sections 5.4–5.6): both errors as functions of line strength, Doppler width, temperature, epoch, wavelength and composition—36 further conditions showing the separation to be universal and, at fixed Doppler width and shell geometry, controlled by the realized maximum line optical depth;

- a **bin-width control** (Section 5.5) showing the expansion-opacity error to be invariant from 0.025 to 41 lines per transport bin, so that it is intrinsic to the formalism rather than an artifact of applying it outside the many-line regime it was built for (F13);

- thirteen **methodological findings** (F1–F13, collected in Table 2), among them an exact symmetry cancellation that makes gradient-based validity diagnostics conservative (F1); the demonstration that expansion opacity attenuates a single strong line by $e^{-(1-e^{-\tau})}$ rather than $e^{-\tau}$ (F4); that this error is *per-resonance*, so it neither averages away with line count (F5) nor vanishes as line widths shrink (F6); and two conventions—thermal emission (F8) and spectral normalization (Section 5.6)—whose mismatch produces spurious few-percent discrepancies in exactly the cross-code comparisons such a study depends on.

Readers with no radiative-transfer background should read

Section 2 first; the appendix derives every formula used in the text at a level that assumes only basic calculus.

2 A COMPACT RADIATIVE-TRANSFER PRIMER

2.1 Light through matter: intensity, opacity, optical depth

Consider a beam of light of frequency ν traveling through a gas. Its strength is described by the *specific intensity* I_ν (energy per unit time, area, solid angle, and frequency). Along a path length s , the gas removes energy from the beam (absorption) and adds its own (emission):

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu. \quad (1)$$

Here α_ν (units cm^{-1}) is the *absorption coefficient*: $1/\alpha_\nu$ is the average distance a photon travels before being absorbed. The emissivity j_ν describes how much the gas glows. In this work the gas is kept in conditions where its own glow at the relevant frequencies is negligible or exactly modeled (Section 3.2), so absorption is the star of the show.

The dimensionless quantity that controls absorption is the *optical depth*

$$\tau_\nu = \int \alpha_\nu ds, \quad (2)$$

which counts how many absorption mean-free-paths the beam crosses. If the gas does not glow, the solution of Eq. (1) is simply

$$I_\nu^{\text{out}} = I_\nu^{\text{in}} e^{-\tau_\nu} : \quad (3)$$

a beam crossing optical depth $\tau = 1$ keeps 37% of its light, $\tau = 2$ keeps 14%, $\tau = 5$ keeps 0.7%. Every result in this paper is, at bottom, a statement about which τ gets exponentiated. (The full solution with emission is derived in Appendix A1.)

2.2 Spectral lines and Doppler width

Atoms absorb light most strongly at the discrete frequencies of their internal transitions—*spectral lines*. A line is not infinitely sharp: thermal motion of the absorbing ions Doppler-shifts each ion’s absorption slightly, smearing the line into a profile $\phi(\Delta\nu)$ (normalized so $\int \phi d\nu = 1$) whose width corresponds to a velocity

$$v_D = \sqrt{2k_B T / m_{\text{ion}}}, \quad (4)$$

typically $\sim 0.6 \text{ km s}^{-1}$ for lanthanum at kilonova temperatures. Expressed as a fraction of the line frequency this is $v_D/c \sim 2 \times 10^{-6}$: lines are extraordinarily narrow. The absorption coefficient of one line is

$$\alpha_\nu = \sigma_{\text{cl}} f n_l \phi(\nu - \nu_0), \quad \sigma_{\text{cl}} \equiv \frac{\pi e^2}{m_e c} = 0.02654 \text{ cm}^2 \text{ Hz}, \quad (5)$$

where ν_0 is the line’s rest frequency, f its dimensionless *oscillator strength* (how intrinsically strong the transition is), and n_l the number density of ions sitting in the transition’s lower energy level. How many ions sit in which level is set, in thermal equilibrium, by the Boltzmann distribution (Appendix A1).

2.3 Expanding ejecta: why geometry rescues us

Kilonova ejecta expand *homologously*: after a short time every parcel coasts at constant speed, so its position is $r = vt$ and equivalently

$$v(r) = r/t, \quad (6)$$

where t is the time since the merger. Velocity increases linearly outward, like raisins in an expanding loaf.

Now follow a single photon of fixed frequency ν flying through this medium. Each gas parcel it passes sees the photon Doppler-shifted by the local line-of-sight velocity. Because the velocity changes monotonically along the flight path, the photon's frequency *as seen by the gas* sweeps continuously redward. It therefore comes into resonance with any given line at **one specific location** — the point where the local Doppler shift brings the gas-frame frequency onto the line center — and is out of resonance everywhere else. In homologous flow this location is particularly simple: for a photon traveling toward an observer along the z axis, the line-of-sight velocity of the gas at coordinate z is z/t for *every ray*, so the resonance locations are flat planes $z = \text{const}$.

2.4 The Sobolev approximation

Sobolev (1960) realized that this geometry makes the line optical depth computable without resolving the profile at all; Castor (1970) cast the same idea in the escape-probability form used by most modern codes. The photon crosses the narrow resonance region quickly; if the gas properties (n_l , temperature) are constant across that thin region, the integral in Eq. (2) collapses (Appendix A2) to the *Sobolev optical depth*

$$\tau_S = \sigma_{cl} f n_l(r_{\text{res}}) \lambda_0 t \quad (7)$$

where λ_0 is the line's rest wavelength and n_l is evaluated *at the resonance point*. Two assumptions are buried here and both are testable: (i) *locality* — the gas does not change across the resonance region (width $\sim v_D$ in velocity); (ii) *isolation* — no other line's resonance region overlaps this one. A photon crossing k line resonances is attenuated by $e^{-\sum_k \tau_{S,k}}$.

2.5 Expansion opacity: the workhorse shortcut

Transport codes need an opacity per frequency *bin*, not per line. The *expansion opacity* formalism (Karp et al. 1977; Eastman & Pinto 1993) assigns to each bin an effective absorption coefficient built from the Sobolev depths of all the lines inside it,

$$\alpha_{\text{bin}}^{\text{exp}} = \frac{1}{ct} \sum_{k \in \text{bin}} \frac{\nu_k}{\Delta \nu_{\text{bin}}} (1 - e^{-\tau_{S,k}}). \quad (8)$$

This is enormously convenient, and it is what a code means by “the Sobolev approximation” whenever it works from binned opacities rather than from individual line interactions. But notice what it does to a single line: integrating α^{exp} over the path on which that line stays inside one bin gives an effective optical depth of $1 - e^{-\tau_S}$, so the transmitted fraction is

$$e^{-(1-e^{-\tau_S})} \quad \text{instead of the true} \quad e^{-\tau_S}. \quad (9)$$

For a weak line ($\tau_S \ll 1$) these agree, since $1 - e^{-\tau} \approx \tau$. For a strong line they do not: at $\tau_S = 2$ the true transmission

is 0.135 while expansion opacity gives 0.421; at $\tau_S = 50$, the true transmission is essentially zero while expansion opacity still leaks $e^{-1} = 0.37$ per crossing. Each resonance crossing is “capped” at one effective optical depth, no matter how black the line really is (derivation in Appendix A4). **This substitution, $\tau \rightarrow 1 - e^{-\tau}$ per resonance, is the central object measured in this paper.**

3 METHODS: THE THREE-WAY HARNESS

3.1 Design principle

The danger in comparing two radiative-transfer treatments is that differences can hide anywhere: different atomic data, different level populations, different geometry, different frequency grids. Following the design requirement that only the line-transfer treatment may differ, every experiment in this paper evaluates *one* physical configuration three ways:

Agreement of the first three (analytic, deterministic, SEDONA resolved) validates the harness; the offset of the fourth is the measured error, reported as

$$\Delta_{\text{exp}} = \frac{F_{\text{exp}} - F_{\text{resolved}}}{F_{\text{resolved}}}, \quad \Delta_{\text{Sob}} = \frac{F_{\text{Sob}} - F_{\text{resolved}}}{F_{\text{resolved}}}, \quad (10)$$

where F is the band-averaged transmitted flux. The two differ in an important way. Δ_{exp} is always formed from the *two* SEDONA runs, so Monte Carlo conventions, frequency grids and normalizations cancel identically; the deterministic solver serves as an independent check of the resolved leg. Δ_{Sob} cannot be formed that way, because SEDONA has no per-line Sobolev mode: it necessarily compares an analytic leg against a SEDONA run, and is therefore exposed to three conventions that must be matched by hand—thermal emission (§3.6), spectral normalization (§5.6) and transport treatment (Appendix A7). Each of the three has produced a spurious few-percent result in the course of this work, and one of them produced a retracted one (§5.3). Same-code differentials are the robust ones; cross-code differentials are only as good as their shared normalization.

3.2 Geometry and physical conventions

All experiments use a spherical, homologously expanding shell (Eq. 6) around a central blackbody “light bulb” of radius r_{core} and temperature T_{core} : the shell imprints absorption lines on the core's smooth continuum, and the emergent spectrum is measured far away. Rays that hit the core start with its blackbody intensity; rays that miss it start dark. Table 1 gives the numerical configuration in full.

Three deliberate simplifications keep the comparison interpretable:

(i) **Pure absorption, local thermal emission.** Scattering and fluorescence are off; absorbed energy is re-emitted thermally with source function $S_\nu = B_\nu(T)$ (the Planck function). Shell temperatures are chosen so the shell's own glow in the measured band is negligible ($B_\nu(T_{\text{shell}}) \ll B_\nu(T_{\text{core}})$); every leg treats the identical source, so any residual emission cancels in Δ_{exp} .

Leg	Role
Analytic: Sobolev theory, Eq. (7), <i>p</i> -averaged over the source disk (App. A5)	prediction
Deterministic: <code>formal_transfer.py</code> (this work), frequency-resolved formal solution along rays	resolved reference
Monte Carlo: SEDONA (Kasen et al. 2006; Roth & Kasen 2015) (a) resolved bound–bound mode	resolved cross-check
(b) expansion-opacity mode	under test

Table 1. Numerical configuration of the production runs. The values are identical across every leg and every comparison unless a section states otherwise; the sweeps vary only $\tau_{\max}$ (through the ion density), v_D , T_{shell} and the epoch.

Quantity	Value
Core radius r_{core}	8.64×10^{12} cm
Outer radius r_{out}	2.592×10^{13} cm ($3r_{\text{core}}$)
Epoch t	86400 s (day 1)
Velocity span Δv_{shell}	1000–3000 km s $^{-1}$
Core temperature T_{core}	6000 K
Core luminosity	6.894×10^{37} erg s $^{-1}$
Shell temperature T_{shell}	3000 K (2500–5000 K on the T axis)
Fiducial Doppler width v_D	100 km s $^{-1}$ (1–300 km s $^{-1}$ swept)
Transport grid	$\nu \in [7.50, 7.95] \times 10^{14}$ Hz, $\Delta\nu/\nu = 4.17 \times 10^{-5}$
Spectrum grid	$\Delta\nu/\nu = 10^{-4}$
Band averaged	3800–3955 Å
Normalization margin	3952–3970 Å (line-free)
SEDONA packets	2×10^6 per run
Solver impact parameters	150 rays
Solver frequency points	1600
Electron scattering	off ($\epsilon = 1$)
Radiative equilibrium	disabled

(ii) **Fixed temperatures.** Radiative equilibrium is disabled; the gas state never feeds back on the radiation. Populations are therefore known exactly.

(iii) **Controlled velocities.** Shell velocities span 1000–3000 km s $^{-1}$ rather than the realistic 0.1–0.3c. This keeps the difference between SEDONA’s frozen steady-state transport and a worldline-consistent treatment negligible for the present measurements (F3/F11, Appendix A7), and keeps the frequency-grid cost of resolving line profiles affordable.

3.3 Population control

SEDONA computes its own thermal-equilibrium level populations from the atomic data file it is given. To guarantee both codes use *exactly* the same populations, we generate SEDONA’s atomic-data files directly from the same source data the Python side reads: for the La II experiments, all 472 energy levels of the GSI La II model are written into the file, only the transitions in the studied wavelength window are included as lines, and the ionization potential is set to 10^5 eV so that ionization is impossible and the element sits entirely in the intended ionization stage. Both codes then evaluate the same Boltzmann distribution over the same levels; we verified in SEDONA’s output gas state that the free-electron fraction is $\sim 10^{-10}$, i.e. the population control holds.

3.4 The deterministic reference solver

The independent resolved leg is a deliberately small (~ 200 -line) Python program. It casts parallel rays through the sphere at many impact parameters p , evaluates the frequency-resolved line opacity (Eq. 5) at every step using the local Doppler-shifted frequency, and accumulates the formal solution of Eq. (1) by cumulative optical depth (Appendix A1). Emergent luminosity is the p -weighted integral of ray intensities. Its correctness rests on four analytic limits enforced as unit tests: a bare core reproduces the exact blackbody-sphere luminosity; a single line’s absorption trough reaches $e^{-\tau_s}$ in the Sobolev limit; two well-separated lines absorb independently while overlapping ones multiply ($e^{-(\tau_1+\tau_2)}$); and a warm shell partially refills the trough. The full test suite of the accompanying repository (68 tests) passes at every commit reported here.

3.5 The Monte Carlo noise floor, measured

Every number quoted from SEDONA carries sampling noise, and since several results below are at the sub-percent level that noise has to be measured rather than asserted. We reran identical configurations with different random-number seeds, holding model, atomic data, grid and packet count fixed, so that the spread is pure sampling error. Five seeds at $(\tau_{\max}, v_D) = (5, 100 \text{ km s}^{-1})$ give a fractional standard deviation of 0.26% on the resolved band flux and 0.30% on the expansion band flux; five seeds at $v_D = 10 \text{ km s}^{-1}$ give 0.30%.

That the two widths agree to within 15% of each other matters. The band flux is a sum over all bins in the window, so its sampling error is set by the number of packets landing in the band rather than by how finely the band is subdivided, and should therefore be independent of v_D at fixed `core_n_emit`. Measuring at two widths an order of magnitude apart tests that expectation instead of assuming it, which is what licenses quoting the same uncertainty at the $v_D = 1 \text{ km s}^{-1}$ frontier point, where repeating the run five times would cost some sixty times more.

We therefore attach $\pm 0.3\%$ (1σ) to every band flux, and $\pm 0.3\%$ to Δ_{Sob} , in which only the SEDONA reference fluctuates (the Sobolev leg being analytic). For Δ_{exp} , adding the two SEDONA legs in quadrature gives $\pm 0.4\%$, which is an upper bound: the two runs share a model and differ only in opacity treatment, so their sampling errors are partially correlated. One consequence is worth stating plainly, because it cuts against us: the production runs quoted throughout are single realizations, and the headline resolved band flux (0.3435) sits about 2σ above the five-seed mean (0.3419).

3.6 A convention that must be matched: thermal emission

Finding F8, recorded because it is a trap for anyone benchmarking a deterministic solver against a Monte Carlo code. Our solver integrates the LTE source term $S_\nu = B_\nu(T_{\text{shell}})$ along every ray; SEDONA, run at fixed temperature with radiative equilibrium disabled, deposits absorbed packet energy without re-emitting it. Setting $T_{\text{shell}} \rightarrow 0$ in the solver isolates the difference exactly: the single-ion forest moves $0.3587 \rightarrow 0.3439$ against SEDONA’s 0.3435 (+4.4% $\rightarrow$ +0.1%), and the blend $0.2410 \rightarrow 0.2249$ against 0.2277 (+5.8% $\rightarrow$ -1.2%). Like for like, the two independent codes agree to 0.1% on the single-ion forest—inside the 0.3% Monte Carlo noise of §3.5, so the residual is not resolvable—and to 1.2% on the blend. The term exceeds the naive $B_\nu(3000)/B_\nu(6000) = 2 \times 10^{-3}$ estimate because the shell subtends nine times the core’s projected area, and it grows with saturation. Absolute band fluxes quoted from SEDONA in this configuration are therefore attenuation-only; Δ_{exp} , a differential between two SEDONA runs sharing one convention, is unaffected.

3.7 Atomic data

Real-line experiments use the GSI Database for Kilonova Radiative Transfer (Flörs et al. 2026b,a). It is one of several recent efforts to supply r-process opacity data (cf. Tanaka et al. 2020, 2025); we adopt it because its wavelengths are calibrated against experiment, which matters when the physical question turns on separations of a few km s^{-1} . It provides calibrated energy levels and E1 transitions for singly and doubly ionized lanthanides, with per-level calibration flags (`xmatch` = matched to experiment, `shifted` = calibrated shift, `uncalib`). For La II: 472 levels and 17,743 transitions. Internal consistency of the dataset is excellent: oscillator strengths recomputed from the Einstein A coefficients agree with the tabulated $\log gf$ values to 1.5×10^{-4} across our window, so both codes carry identical line strengths regardless of which column they read.

4 VERIFICATION EXPERIMENTS

4.1 Reproducing and breaking the Sobolev limit in isolation

Before any code comparison, the toy problem: one line, constant n_l , homologous slab. The resolved integral (Eq. 2) converges to the analytic τ_S (Eq. 7) to machine precision ($E_{\text{Sob}} = 2 \times 10^{-16}$ at 2×10^5 grid points), with monotone convergence under refinement—important because an unconverged grid mimics a physical Sobolev deviation.

Breaking locality on purpose (Fig. 1) produced **Finding F1**: with the population gradient placed *symmetrically* about the resonance point, the error cancels identically—the line profile is even while the leading gradient term is odd, and the natural symmetric choice (tanh centered on resonance) also has vanishing curvature there. The error only appears when the resonance sits off-center on the gradient (we use the point where curvature is large), and then follows a clean ϵ^2 law, at the few-percent level when $\epsilon = v_D/v_{\text{scale}} \gtrsim 0.3$ (Appendix A3). The practical consequence: widely used validity diagnostics based on the gradient magnitude $G =$

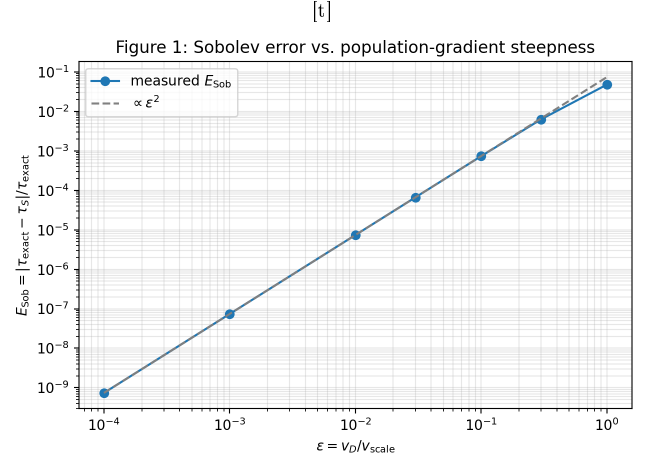


Figure 1. Controlled breakdown of the Sobolev locality assumption. A tanh-shaped population gradient of velocity scale v_{scale} is imposed; the relative error E_{Sob} between the resolved integral and the Sobolev formula follows the predicted ϵ^2 power law in $\epsilon = v_D/v_{\text{scale}}$ (measured log-slope 2.0000), reaching $\sim 5\%$ when the gradient scale approaches the Doppler width.

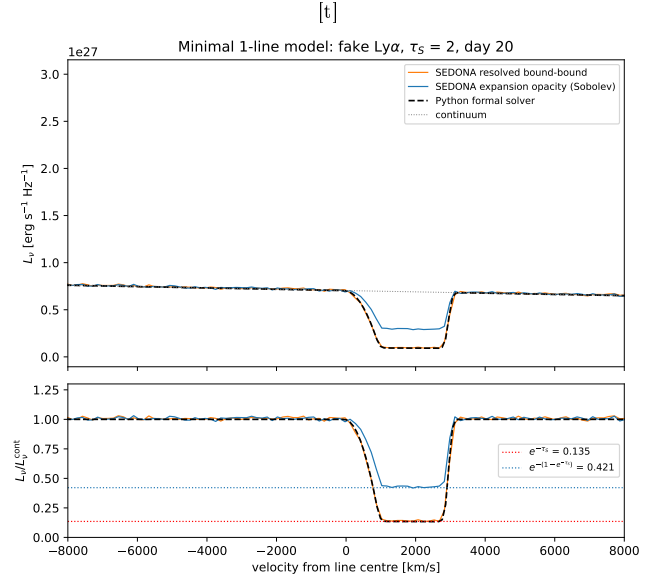


Figure 2. Minimal one-line model (artificial Ly α -like line, $\tau_S = 2$, cold neutral shell). Top: emergent spectra. Bottom: ratio to the continuum. SEDONA’s resolved mode (orange) lies on top of the deterministic solver (black dashed) across the entire profile, at the analytic depth $e^{-\tau_S} = 0.135$ (red dotted). The expansion-opacity mode (blue) instead sits at the predicted capped value $e^{-(1-e^{-\tau_S})} = 0.421$ (blue dotted).

$v_D |d \ln n_l / dv|$ are *conservative*: the true error depends on where the resonance sits on the gradient, and can be far below the G -based bound.

analytic $e^{-\tau_S}$	0.1353
deterministic solver	0.1372
SEDONA resolved	0.1420
SEDONA expansion opacity	0.4285
predicted cap $e^{-(1-e^{-\tau_S})}$	0.4212

analytic Sobolev (p -averaged)	0.2405
deterministic solver	0.2407
SEDONA resolved	0.2435
analytic expansion cap	0.3194
SEDONA expansion opacity	0.3257

4.2 One line, three codes

The minimal cross-code experiment uses SEDONA’s built-in two-level test atom (one line at 1215.5 Å, $f = 0.665$) in a shell tuned to $\tau_S = 2$. The shell temperature (2000 K) was chosen after an ionization- balance check: at the very low densities that give $\tau_S \sim 1$, a 5000 K gas would be fully ionized, destroying population control; at 2000 K the gas is neutral to one part in 10^{10} .

The measured absorption-trough depths (Fig. 2):

Two independent resolved implementations—deterministic rays and Monte Carlo—agree with each other and with theory at the few-percent level (**harness validated**), while the expansion-opacity mode lands within 2% of its own predicted cap, a factor 3.2 too much transmitted light. This is **Finding F4**: for a single strong line, “Sobolev mode” as implemented via expansion opacity is *not* Sobolev line transfer.

The same experiment surfaced **Finding F3**: the deterministic solver’s 1.4% excess over the analytic value scales with the bulk velocity at the resonance, and at a resonance placed at 7.7% of c grows to 7%, matching $e^{-\tau_S(1-z_{\text{res}}/ct)}$ to four digits. We read this first as a frame ambiguity and then as the leading relativistic correction; both were wrong. Appendix A7 shows it is an artifact of integrating a *frozen snapshot* of the ejecta: allowing the ejecta age to advance along the photon path, as it physically must, gives $\tau_{\text{rel}}/\tau_S = 1/\gamma$, which has no first-order term at all. The practical consequence is the opposite of what the $\mathcal{O}(v/c)$ reading implied: at our shell velocities the relativistic correction is 5×10^{-5} , so it cannot account for any part of the residual in §5.3.

4.3 From two lines to a twenty-line ladder

Custom SEDONA atomic files with 2 and then 20 identical-strength lines ($\tau_S = 0.5$ each) test multi-line behavior (Fig. 3). In the center of the 20-line forest, where each photon crosses ~ 3 resonances, five determinations of the transmitted fraction give:

Finding F5: the expansion-opacity error tracks its own per-line cap prediction, i.e. the error is incurred *per resonance crossing*. Adding more lines multiplies both the true and the capped attenuations equally, so the fractional flux error does *not* average away with line count—it only vanishes as $\tau \rightarrow 0$ per line. Expansion opacity is a *weak-line* approximation, not a many-line one.

A technical note that matters for anyone reproducing this: the analytic staircase must be averaged over the stellar disk (p -average, Appendix A5), because resonance planes closer than the core radius are crossed by only the outer rays; naive plane counting predicts visibly too-shallow troughs.

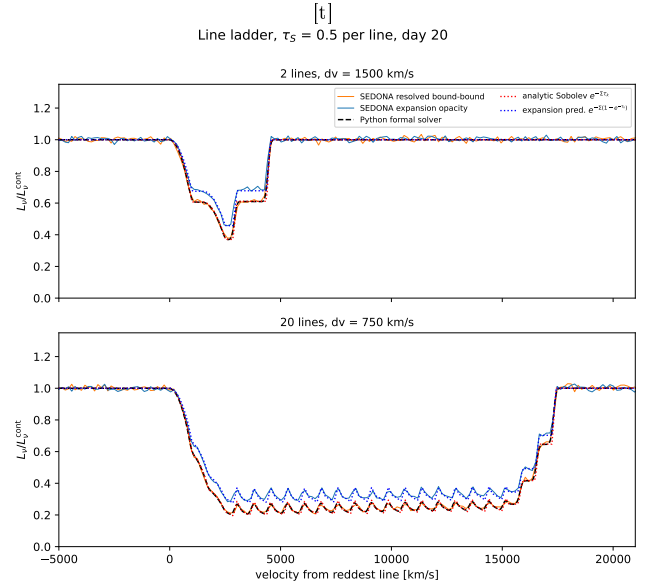


Figure 3. Line ladder at $\tau_S = 0.5$ per line. Top: two lines separated by 1500 km s^{-1} (overlapping troughs). Bottom: twenty lines spaced 750 km s^{-1} (a mini forest). The resolved trio—SEDONA resolved (orange), deterministic solver (black dashed), p -averaged analytic staircase (red dotted)—lock together tooth for tooth; the expansion pair (blue solid: SEDONA; blue dotted: analytic cap prediction) ride systematically above them.

5 RESULTS: TWO APPROXIMATIONS, TWO ERRORS

With the harness validated, we turn to calibrated La II data. We first ask whether the assumptions underlying either approximation have anything to fear in real line lists (§5.1), then set up a realistic forest (§5.2) and measure both errors on it (§5.3)—the paper’s central result. The remaining subsections establish where each error lives (§5.4), that the picture is not peculiar to one window or epoch (§5.6), how blending changes it (§5.7), and what the shortcut buys in computer time (§5.8).

5.1 Do the assumptions have anything to fear? Line crowding

The geometric isolated-line condition is violated when profiles crowd within a Doppler width; whether that violation produces a *transport* error depends on the interaction physics, and §5.3 shows it does not in the pure-absorption case studied here, where optical depths add exactly. Fig. 4 shows that even for this *single* ion the calibrated data contain a real crowding tail: 11.3% of the broader line sample has a neighbor within 10 km s^{-1} . Multi-ion mixtures only add lines, so the geometrically crowded regime exists in nature—which is what makes it worth establishing that crowding alone is not sufficient for a transport error.

5.2 A realistic forest: 153 calibrated lines

We select the 100 Å window with the richest optical-depth distribution ($3850\text{--}3950 \text{ Å}$), build the population-controlled SEDONA atom from the GSI data (Section 3.3), and run the

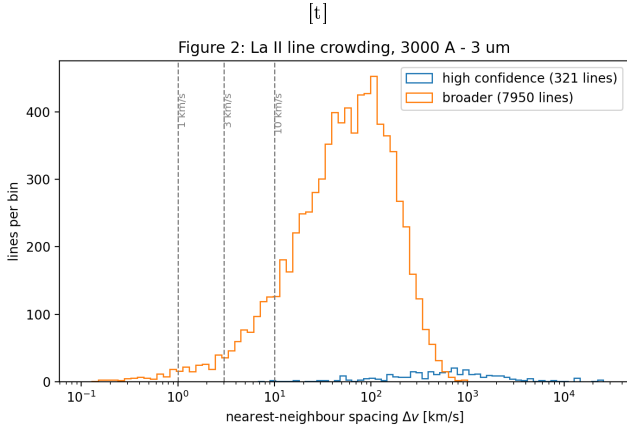


Figure 4. Nearest-neighbor velocity spacing of GSI La II lines (3000 Å–3 μm): in the broader subset, 11.3% of lines have a neighbor within 10 km s^{−1} and 3.3% within 3 km s^{−1}.

deterministic solver, $S_\nu = B_\nu(T_{\text{shell}})$	0.3587
deterministic solver, $T_{\text{shell}} \rightarrow 0$	0.3439
SEDONA resolved	0.3435
SEDONA expansion opacity	0.4973
Δ_{exp}	+44.8%

three-way comparison at $T = 3000$ K, $t = 1$ day, Doppler width 100 km s^{−1} (Fig. 5). Band-averaged transmitted flux over 3800–3955 Å:

The resolved pair agree to 4.4% as tabulated—and to 0.1%, well inside the 0.3% Monte Carlo noise of §3.5, once the thermal-emission convention is matched (Finding F8, §3.6)—while the expansion-opacity run overestimates the transmitted band flux by 45%. Notably this exceeds the uniform-ladder error (+33.8% at $\tau = 0.5$ per line): real forests are dominated by a few strong lines, precisely where the per-crossing cap is most wrong. That observation prompts the question the next subsection answers—whether the fault lies with the Sobolev approximation or with the way it has been implemented.

5.3 Which approximation is at fault? Separating the two errors

We now put both approximations to the same forest and the same truth. The distinction drawn in §2.5 becomes operational here: the Sobolev approximation *itself* attenuates each line by $e^{-\tau_S}$ with delta-function resonances, while expansion opacity substitutes $1 - e^{-\tau_S}$ at every crossing. In this pure-absorption LTE configuration the Sobolev prediction is exact analytics (the p -averaged staircase of Appendix A5), so it requires no additional transport calculations, and the identical machinery yields the expansion prediction by damping each crossing. A Monte Carlo realization of per-line Sobolev interactions would add only sampling noise to a result available in closed form.

For the La II forest, against SEDONA resolved = 0.3435: Sobolev proper gives 0.3508 (+2.1%) while expansion opacity gives 0.4973 (+44.8%). Across the sweep (Fig. 6):

Down the width axis at $\tau_{\text{max}} = 5$, where SEDONA resolved

τ_{max}	v_D [km s ^{−1}]	Δ_{Sob}	Δ_{exp}
0.5	10	+0.2%	+3.2%
0.5	300	+0.0%	+2.8%
5	10	−0.2%	+38.5%
5	300	+9.2%	+53.0%
50	10	+1.1%	+45.3%
50	300	+32.9%	+93.8%

v_D [km s ^{−1}]	Δ_{Sob}	Δ_{exp}
1	−0.3%	+39.1%
3	−0.3%	+39.2%
10	−0.2%	+38.5%
30	+0.5%	+41.3%
100	+2.5%	+45.0%
300	+9.2%	+53.0%

runs already existed at the thermal end, the Sobolev error simply switches off while the expansion error does not:

An earlier version of this table reported a v_D -independent Sobolev error of 5–8%. That was a normalization artifact: the sweep normalized SEDONA band fluxes by raw luminosity in a red margin rather than by the continuum ratio, leaving the Planck slope across the band in the answer, and the analytic leg—correctly normalized—was compared against it. The expansion column is essentially unchanged, because it is a same-code differential in which the bias cancels. All comparisons now go through a single shared routine, and a line-free control window is required to return unity.

Finding F9: the two components of F6 have different owners. (F6, from §5.4 below, decomposes Δ into a strength-set floor plus a term that grows with v_D .) *The strength-set floor belongs to expansion opacity alone.* Sobolev proper exhibits no strength floor at all—at $v_D = 10$ km s^{−1} it is +0.2%, −0.2% and +1.1% at $\tau_{\text{max}} = 0.5, 5$ and 50 respectively, with no trend in strength—while expansion climbs +3.2% → +38.5% → +45.3% across the same three. That floor is the per-crossing cap, not a failure of locality or isolation. *The width-growing term, by contrast, is not a strength effect at all:* a delta-function resonance has no width by construction, so the Sobolev leg is exactly v_D -independent, and the resolved calculation's width dependence appears as clipping of line profiles against the boundaries of the finite line-forming region (Appendix A6). It reaches +32.9% at $(\tau_{\text{max}}, v_D) = (50, 300 \text{ km s}^{-1})$ —where v_D is 15% of the shell's velocity span, a configuration with no counterpart in real ejecta, for which $v_D/\Delta v_{\text{shell}} \sim 10^{-4}$.

The headline is therefore sharper than either treatment alone suggests: at kilonova-relevant line widths ($v_D \leq 30 \text{ km s}^{-1}$) **the Sobolev approximation is accurate to $\lesssim 2\%$ across the full strength range tested—and to $\lesssim 0.5\%$ at $\tau_{\text{max}} = 5$, the case followed down to 1 km s^{−1}—while its standard expansion-opacity implementation errs by +38–48%.** The literature conflates the two under one name; they are different approximations and they fail for different reasons.

5.3.0.1 Overlap is not the residual, and cannot be.

It is tempting to attribute the remaining Sobolev error at broad v_D to the isolation assumption, given the line crowding documented in §5.1. In this harness that is impossible. With fixed populations and pure absorption the opacities add, so

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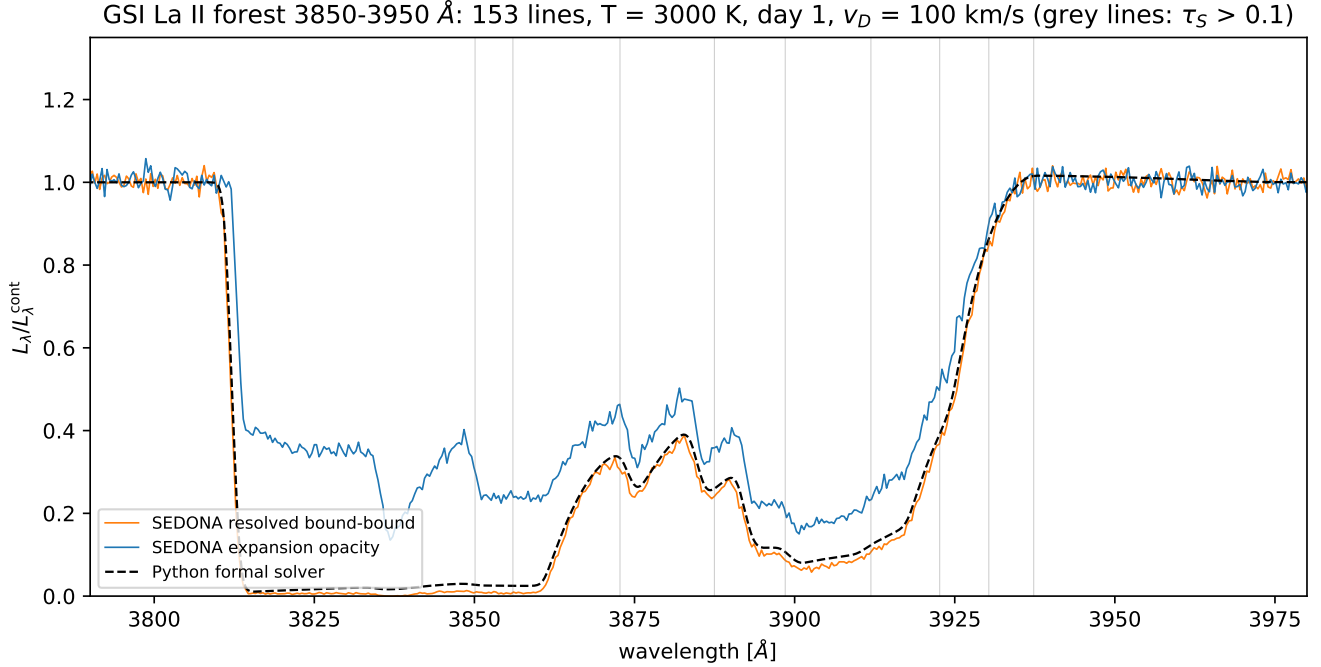


Figure 5. The 3850–3950 Å La II forest (153 calibrated lines, 3 with $\tau_S > 1$, max 5) at $T = 3000$ K, one day after merger. The resolved treatments (orange, black dashed) saturate to black where strong troughs overlap; expansion opacity (blue) never drops below ~ 0.2 because each crossing is capped.

$\tau_\nu = \sum_i \int \alpha_{\nu,i} ds = \sum_i \tau_{\nu,i}$ *exactly*, and the emergent intensity is $I_\nu = I_{\nu,0} \exp(-\sum_i \tau_{S,i})$ whether or not the profiles overlap. Two identical lines driven from 20 Doppler widths apart to exact coincidence reproduce the Sobolev prediction to six decimal places. Line overlap becomes a genuine effect only once scattering, fluorescence or NLTE populations let one transition modify the radiation another one sees—all switched off here by design. The isolation assumption simply cannot be probed by an attenuation-only calculation.

The residual is instead *geometric*, and Appendix A6 derives it: where a resonance lies within a few Doppler widths of a boundary of the line-forming region, the resolved profile is clipped while Sobolev applies a hard step, giving $\tau/\tau_S = \frac{1}{2}[\text{erf}((z_{\text{hi}} - z_{\text{res}})/\Delta) - \text{erf}((z_{\text{lo}} - z_{\text{res}})/\ell_S)]$ with the Sobolev length $\ell_S = v_D t$. The affected fraction of the band is $\sim v_D/\Delta v_{\text{shell}}$, which is why the error is invisible at thermal widths and reaches tens of percent only when v_D is a sizeable fraction of the velocity span of the shell.

5.4 Where each error lives: strength, width, temperature

The separation just established is a single point in parameter space. We now map where each error lives, by scaling the two quantities the two approximations should respond to differently: the overall line strength, which drives the per-crossing cap, and the Doppler width, which the Sobolev treatment ignores by construction.

Scaling the overall line strength ($\tau_{\text{max}} \in \{0.5, 5, 50\}$ by scaling the ion density) and the Doppler width ($v_D \in$

τ_{max}	v_D [km s ⁻¹]					
	1	3	10	30	100	300
0.5	–	–	+3.2	+2.4	+3.2	+2.8
5	+39.1	+39.2	+38.5	+41.3	+45.0	+53.0
50	–	–	+45.3	+48.1	+60.9	+93.8

$\{1, 3, 10, 30, 100, 300\}$ km s⁻¹) yields the first rows of the validity map (Figs. 7–8; Δ_{exp} in percent):

Finding F6: the error decomposes into

(i) a *strength-set floor*, independent of v_D : the pure per-crossing substitution error. Weak forests are safe at the 3% level; by $\tau_{\text{max}} = 5$ the floor is $\sim 39\%$;

(ii) a *wing term* growing with v_D : expansion opacity is width-blind (its transmitted flux is flat across a factor 300 in v_D), while genuinely wide lines absorb continuum between resonances.

The frontier measurement at 1 km s⁻¹—within a factor 2 of the actual La thermal width—confirms that the floor survives: *no refinement of line widths rehabilitates expansion opacity in the strong-line regime*.

The temperature axis (Fig. 8, left), with strength pinned, moves Δ_{exp} only from +41.8% to +48.1% across 2500–5000 K: in this window the same few strong lines dominate at all temperatures, so population redistribution is a weak axis (single-ion result; multi-ion mixtures may differ).

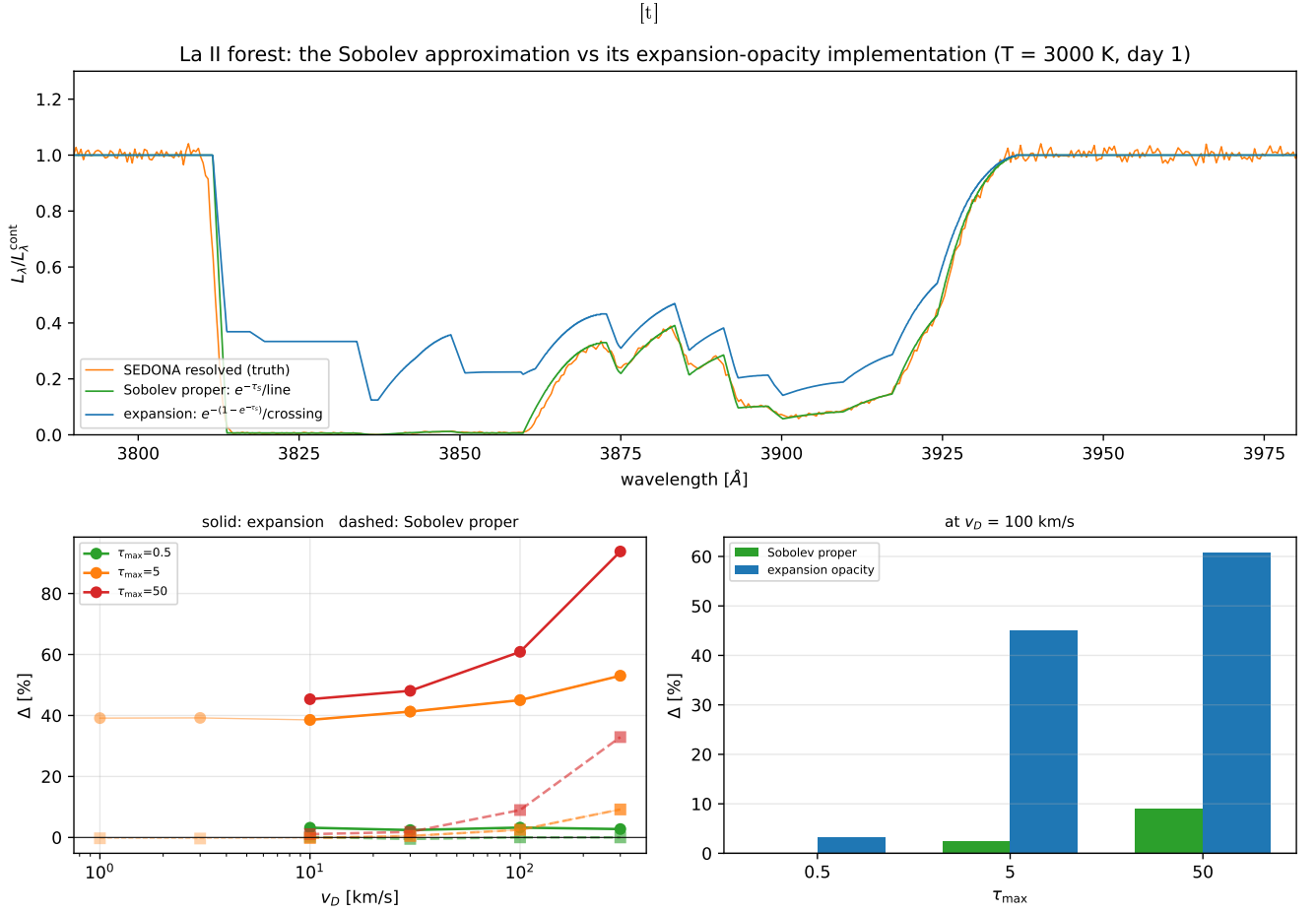


Figure 6. Top: the La II forest under three treatments. The Sobolev approximation proper (green, per-line $e^{-\tau_S}$) tracks the resolved calculation (orange) closely; expansion opacity (blue) sits well above it. Bottom left: both errors versus Doppler width, for three line-strength scalings (solid: expansion; dashed: Sobolev). Bottom right: the two at $v_D = 100 \text{ km s}^{-1}$.

line list	lines/bin	F_{resolved} spread	$\Delta_{\text{expansion}}$
La II (153 lines)	0.025–2.5	0.13%	$+43.7\% \pm 1.2$
+ Ce II (2529)	4.1–41	0.5%	$+15.3\% \pm 0.7$

5.5 Is the error an artifact of bin choice?

Expansion opacity is derived for frequency bins containing *many* lines (Karp et al. 1977; Eastman & Pinto 1993); the statistical averaging is what gives the construction its meaning. Our transport grids are much finer than that— 12.5 km s^{-1} bins against a mean La II line spacing of $\sim 50 \text{ km s}^{-1}$, i.e. about one line per four bins—so it is fair to ask whether we have measured the formalism outside its design regime.

Appendix A4 answers this analytically: the bin width cancels, and each resonance crossing contributes $1 - e^{-\tau_S}$ at any resolution. The numerical test confirms it. Sweeping the transport resolution over two decades at fixed physics (Fig. 9):

The two line lists give different errors—the blend is denser and partly blanketed, which is F7—but neither error depends on the bin width, including at 41 lines per bin, which is squarely inside the many-line regime the construction targets. The error is intrinsic to the formalism.

One methodological point deserves stating because it nearly produced the opposite conclusion. Extending the sweep to bins several times wider than the line profile appears to make $\Delta_{\text{expansion}}$ collapse and change sign. It does not: past that point the *resolved* calculation is no longer resolving anything, its band flux climbing from 0.342 to 0.679, and the comparison is measuring the reference rather than the approximation. A verdict computed over all resolutions without that check reports “not invariant”. The reference’s own convergence has to be established before its disagreement with anything else can be interpreted.

5.6 Is the separation universal? A breadth sweep

The measurements above rest on one wavelength window at one epoch. To test their generality we ran a sweep over four windows (4300, 4900, 7000 and 9100 Å, selected automatically by optical-depth richness and deliberately excluding the window used so far), three epochs ($t = 0.5, 1$ and 3 days at fixed ejecta mass, so $\rho \propto t^{-3}$ and $\tau_S \propto t^{-2}$), and three ion mixtures (La II; +Ce II; +Ce III). That is 36 conditions and 72 transport calculations, spanning 71 to 2504 lines per window and realized τ_{max} from 8×10^{-5} to 34.

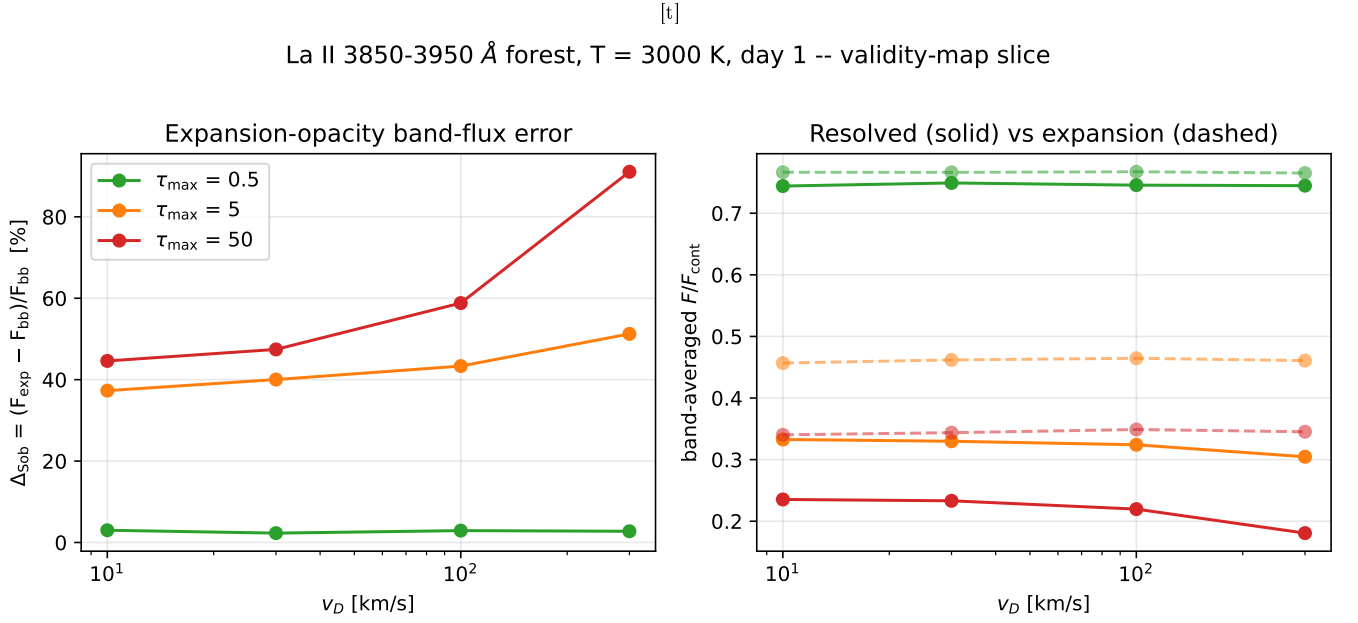


Figure 7. First validity-map slice: Δ_{exp} vs Doppler width for three overall line-strength scalings of the La II forest. Right panel: expansion-opacity fluxes (dashed) are blind to v_D , while resolved fluxes (solid) drop as profiles widen.

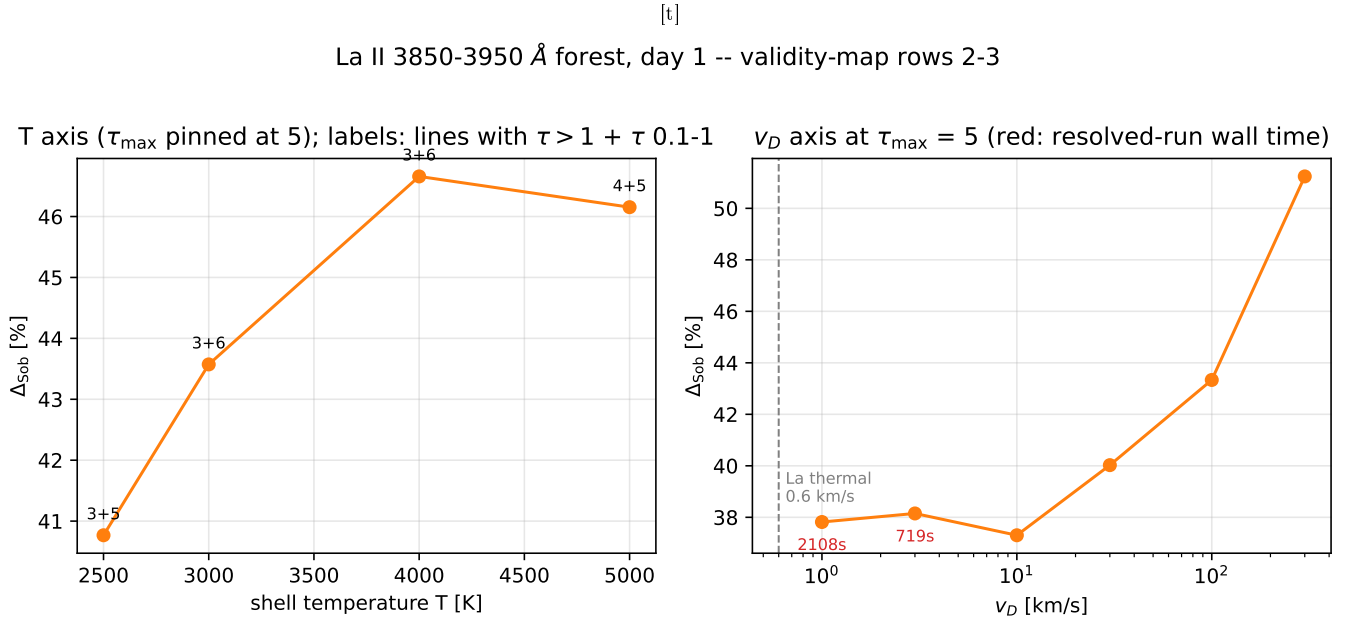


Figure 8. Left: temperature axis at pinned $\tau_{\text{max}} = 5$ (weak dependence; annotations count strong lines). Right: the Doppler-width frontier: Δ_{exp} is flat at $\sim 38\%$ from 10 km s^{-1} down to 1 km s^{-1} , essentially the La thermal width; red annotations give the resolved-run wall time.

	median	max	$\tau_{\text{max}} > 3$	$\tau_{\text{max}} < 0.5$
Δ_{exp}	+5.1%	+61.3%	+20.0%	+0.3%
Δ_{Sob}	+3.6%	+11.3%	+6.6%	+0.5%

Finding F10: the separation is universal, and the realized τ_{max} is the controlling variable *among the axes swept*. All 36 conditions fall on one trend (Fig. 10) regardless of

window, epoch or composition; those enter only through the optical depths they produce. The sweep holds the shell geometry and the Doppler width fixed, and §5.4 shows Δ does depend on v_D ; the collapse is therefore onto τ_{max} within a fixed $(v_D, \Delta v_{\text{shell}})$ slice, not a claim that τ_{max} alone determines the error everywhere. Both errors vanish together as $\tau_{\text{max}} \rightarrow 0$ —the weak-line limit both approximations share—and both grow with τ_{max} , but expansion opacity errs by a me-

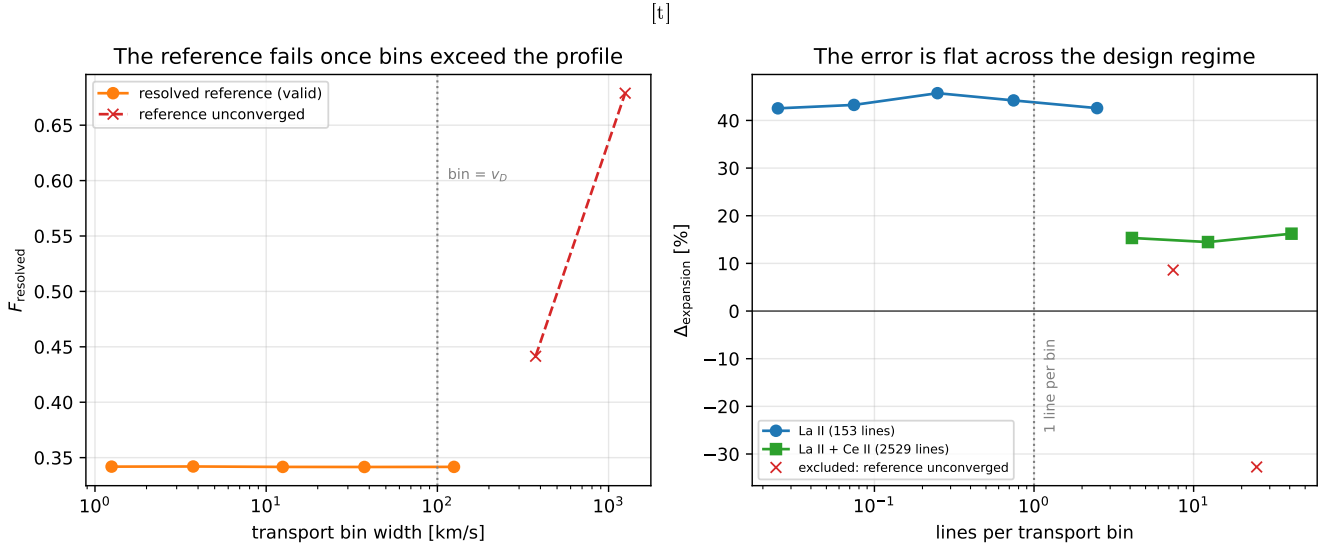


Figure 9. Left: the resolved reference is stable to 0.13% while the transport bin stays narrower than the line profile, and runs away once it does not—those points measure the reference failing, not the approximation changing, and are excluded. Right: across the valid range $\Delta_{\text{expansion}}$ is flat from 0.025 to 41 lines per bin, two line lists and three decades.

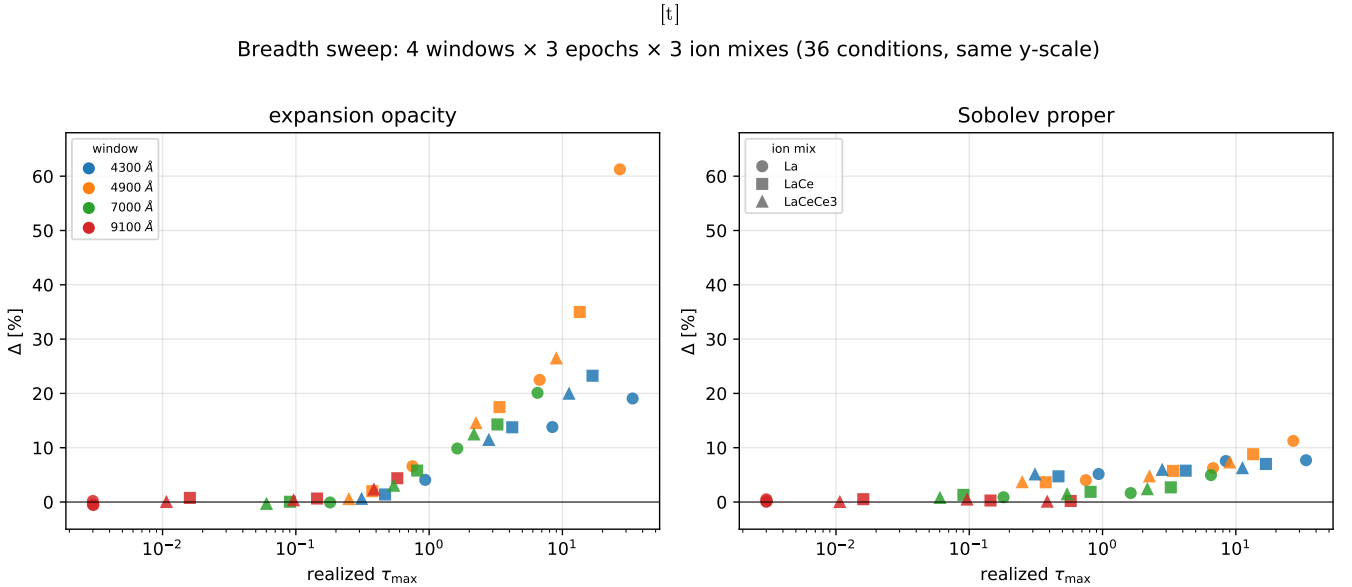


Figure 10. Thirty-six conditions—four wavelength windows (none of them the window of §5.3), three epochs, three ion mixtures—with both panels on the same vertical scale. Every condition collapses onto a single trend in the realized τ_{max} ; expansion opacity reaches +61% while the Sobolev approximation never exceeds +12%.

dian factor 3.5 more than the Sobolev approximation wherever $\tau_{\text{max}} > 1$. This has a practical corollary: the error can be estimated from the strongest line in a band without knowing which ion or epoch produced it.

We note one methodological point that the sweep itself exposed. The 9100 Å window contains no strong La II lines ($\tau_{\text{max}} \simeq 0$) and therefore acts as a free null control: its band flux must be unity. An early pass returned 1.075 instead, tracing to a normalization reference that straddled the final (partial, and unreliable) bin of the transport grid. The

bias cancelled exactly in Δ_{exp} , a same-code differential, but contaminated Δ_{Sob} , which compares a Monte Carlo result against an analytic one—the same asymmetry as the emission convention of §3.6. Null controls of this kind cost nothing and are worth including by default.

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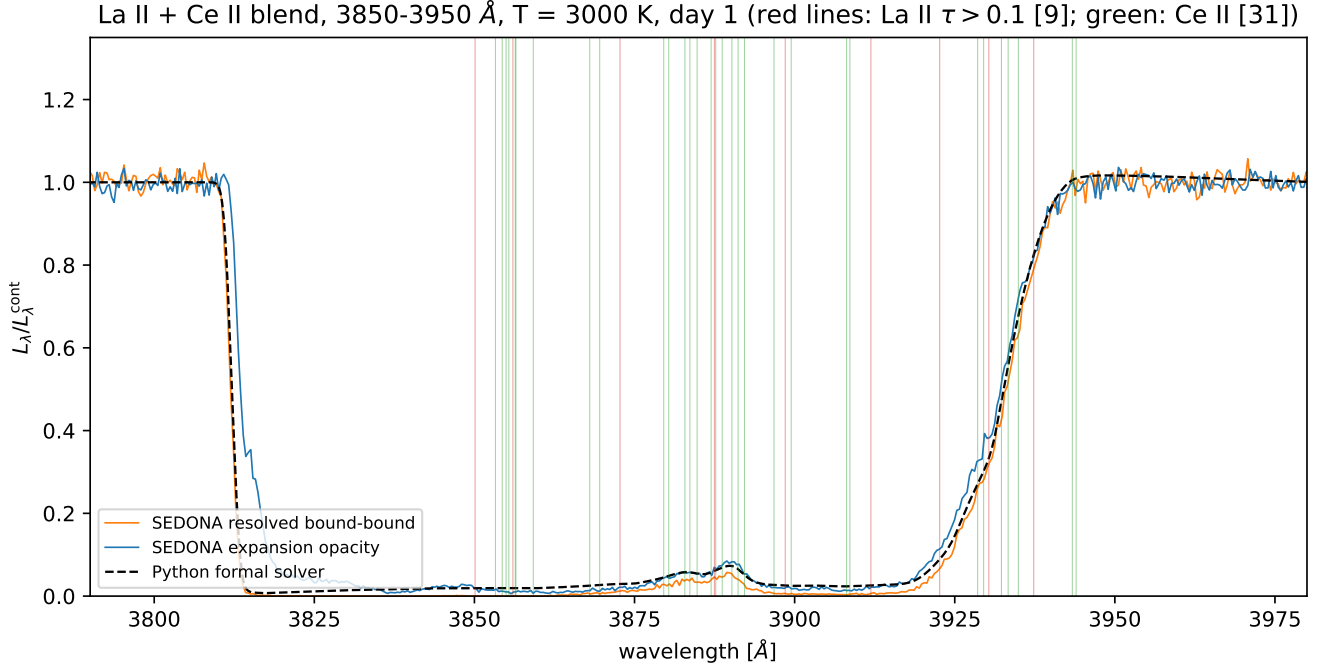


Figure 11. La II + Ce II blend in the same window. Ce floods it with 2376 additional lines (2529 total; strong-line spacings down to 8 km s^{-1}), saturating the band to near-black in every treatment.

5.7 Multi-ion blending and the density dependence

One axis remains: what happens when forests become dense enough that resonances stop being resolvable as individual features?

Adding Ce II at equal mass fraction multiplies the line count by 16 and drives strong-line separations down to 8 km s^{-1} —genuine sub-Doppler blending. The measured error, however, *falls*, to $\Delta_{\text{exp}} = +14.6\%$ (Fig. 11).

Finding F7: Δ_{exp} is non-monotonic in forest density. With enough crossings, even capped per-crossing contributions sum to blackness, so both treatments agree that almost nothing escapes and the relative error is suppressed.

F7 therefore defines the domain of relevance rather than weakening the result. The error is small in two limiting regimes—weak lines, and bands driven completely black—and largest in the *intermediate* one, where optically thick lines coexist with nonzero escaping flux. Partially transparent, feature-forming regions of exactly this kind do occur in realistic kilonova calculations: multi-element models retain identifiable La, Ce and Sr features rather than becoming featureless (Domoto et al. 2022), and r-process opacity varies strongly with composition and by roughly an order of magnitude with epoch, so “lanthanide-rich” does not imply uniform blanketing at all wavelengths and times (Tanaka et al. 2020).

We are explicit about what this does and does not license. It does not license reading +38–48% as the error of a fully mixed kilonova spectrum; F7 shows directly that the error is suppressed as a band approaches complete blanketing, and our La II + Ce II case already demonstrates the suppression. What it does license is the statement that expansion opacity

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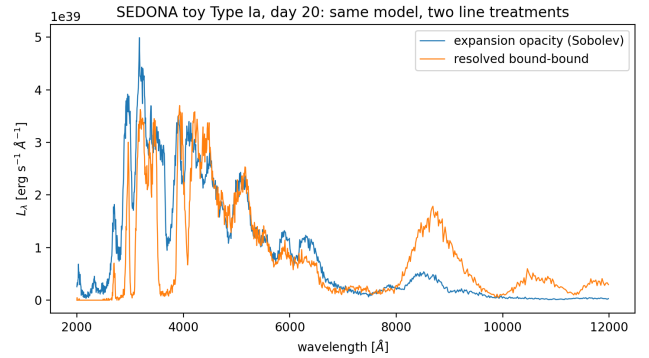


Figure 12. Motivating cost anecdote from a standard SEDONA Type Ia supernova example: the same model runs in 14 s with expansion opacity and 88 min resolved (378×), while redistributing flux from the UV to the near-IR.

can generate tens-of-percent attenuation errors in strong, incompletely blanketed bands, and that such bands are not a contrived configuration.

5.8 Computational cost: what the shortcut buys

Why does everyone use the shortcut? Cost. On a standard SEDONA Type Ia example, switching from expansion opacity to the resolved treatment multiplied the run time by 378 (Fig. 12). Our sweep adds a cleaner measurement: in the con-

trolled forest, resolved-run wall time scales as $1/v_D$ (33 s at 100 km s^{-1} , 719 s at 3 km s^{-1} , 2109 s at 1 km s^{-1}), driven entirely by the transport frequency grid—and the expansion run on the same fine grid costs the same (2305 s at 1 km s^{-1}). The expense is the grid, not the physics; narrow windows keep resolved calculations affordable, which is the strategy this program uses.

6 DISCUSSION

6.0.0.1 What the floor means for kilonova modeling.

Lanthanide-rich ejecta are the definition of the strong-line regime: the opacity calculations that motivate this work find bound-bound optical depths in the line-forming region far above unity across the optical and near-infrared (Tanaka et al. 2020, 2025), and our sweep spans τ_{max} up to 50, where the floor is already $\sim 45\%$ and the analytic cap argument (Appendix A4) shows it does not turn over at larger τ . Our measurements say that in that regime the expansion-opacity treatment overestimates band-integrated transmitted flux by tens of percent *as a floor*, independent of how finely line profiles are treated.

6.0.0.2 What the floor does *not* yet mean. One step separates that statement from a claim about kilonova spectra, and it is worth making explicit because it bounds every astrophysical reading of this paper. Radiative equilibrium is disabled throughout (§3.2): SEDONA deposits absorbed packet energy without re-emitting it, so what we measure is the *attenuation of an externally supplied continuum*, not an emergent spectrum. In an energy-conserving calculation the photons that expansion opacity wrongly lets through a strong line would, under resolved transport, have been absorbed and re-emitted at some other frequency. Total luminosity is conserved; only its spectral distribution differs. A +45% error in attenuated band flux is therefore an upper bound on the corresponding error in an emergent spectrum, not a measurement of it.

What survives that step is the *sign*, which is fixed by the per-crossing cap: the substitution can only ever under-attenuate (Appendix A4), so expansion opacity leaks flux out of strong-line bands in one direction, and thermal re-emission returns it redward of where it was lost rather than restoring the band. The *magnitude* in an emergent spectrum does not follow from anything measured here and must be obtained with energy-conserving transport; redistribution is nonlinear, and we decline to predict its outcome. Repeating these measurements with radiative equilibrium enabled is the natural next experiment, and it is a well-posed one: the configuration here differs from an energy-conserving run by a single SEDONA switch.

6.0.0.3 What exactly was measured on the Sobolev side.

Our Sobolev leg is the approximation in its pure form— $e^{-\tau_S}$ per line, resonances treated as planes—evaluated analytically. That evaluation is *exact* rather than approximate, but only because the configuration is pure absorption without scattering: it is precisely that restriction which makes the p -averaged staircase equal to per-line Sobolev transport. A per-line Sobolev scheme inside a scattering

Monte Carlo code—the macroatom treatment of TARDIS-class codes (Lucy 2002, 2003; Kerzendorf & Sim 2014), say—is a different object, and our numbers should not be read as bounding it; extending the harness there is the natural sequel, and is the regime in which the leg would have to be built rather than computed.

Within that scope the attribution is clean. The per-crossing cap (F4, F5) is absent from the Sobolev leg by construction and present in expansion opacity by construction, and the measured errors follow suit. Of the two Sobolev-intrinsic assumptions, neither is stressed here: locality is not, because our populations are constant (and F1 shows it would cancel at leading order even if it were), and isolation cannot be, because in pure absorption optical depths add exactly regardless of overlap. What remains is the geometric boundary effect of Appendix A6, which is a property of the finite line-forming region rather than of the approximation.

6.0.0.4 Limitations. Pure absorption with LTE populations (no scattering, fluorescence, or NLTE); fixed temperatures; controlled shell velocities ($1000\text{--}3000 \text{ km s}^{-1}$) rather than realistic $0.1\text{--}0.3c$ (which needs a worldline-consistent time-dependent treatment, F11); Doppler widths down to 1 km s^{-1} but not the 0.6 km s^{-1} thermal value; two elements and three ionization stages across four windows and three epochs, but no full lanthanide mixture; Monte Carlo noise measured at $0.26\text{--}0.30\%$ per band flux (Section 3.2); and radiative equilibrium disabled, so the reported fluxes are attenuation of an externally supplied continuum rather than an emergent spectrum.

Within that pure-absorption configuration these caveats do not touch the sign or order of magnitude of the central result, whose mechanism is reproduced independently by analytic theory (Appendix A4). We stress what that sentence does *not* say. Whether the per-crossing cap survives radiative redistribution—where a photon absorbed in one line may re-emerge in another, or at another frequency—is not tested here, and no result in this paper bounds it. That is the subject of a companion study.

7 CONCLUSIONS AND NEXT STEPS

We built and validated a harness that isolates the line-transfer treatment in expanding ejecta—three independent calculations agreeing to $\sim 1\%$ once conventions are matched—and used it to measure, on calibrated lanthanide data, two errors that the literature carries under a single name.

The Sobolev approximation itself is accurate to $\lesssim 2\%$ at kilonova-relevant line widths ($v_D \leq 30 \text{ km s}^{-1}$), and to $\lesssim 0.5\%$ at $\tau_{\text{max}} = 5$ down to 1 km s^{-1} , showing no dependence on line strength; its only measured departure is the neglect of finite line width, and that is a property of imposing a finite line-forming region rather than of the approximation (Appendix A6). Its standard expansion-opacity implementation (Karp et al. 1977; Eastman & Pinto 1993) is a different matter: a per-resonance substitution $\tau \rightarrow 1 - e^{-\tau}$ gives it a strength-set error floor of +38–48% in band flux that no refinement of line widths removes, because the substitution is not a discretization but part of the formalism. Across 36 conditions spanning four windows, three epochs and three ion

Table 2. Findings register.

#	Finding	Where
F1	Symmetric gradients cancel the Sobolev error at leading order; gradient-based diagnostics are conservative	§4
F2	Resolved vs expansion cost: 378× on a standard example	§5
F3	The $\mathcal{O}(v_{\text{bulk}}/c)$ offset from comoving Sobolev τ is an artifact of frozen-snapshot integration, not of relativity	§4, App. A7
F4	Expansion opacity transmits $e^{-(1-e^{-\tau})}$ per crossing, not $e^{-\tau}$	§4
F5	That error is per-resonance: it does not average away with line count	§4
F6	Δ_{exp} = strength-set floor + v_D -growing wing term; the floor survives to thermal widths	§5
F7	Δ_{exp} is non-monotonic in forest density: maximal for strong sparse forests, suppressed at full line blanketing	§5.7
F8	Solver-vs-MC offsets trace to the thermal-emission convention, not to profiles or resolution; matched, the codes agree to $\sim 1\%$	§3.6
F9	The strength floor is expansion opacity's alone; the width term is the boundary effect of F12. At $v_D \leq 30 \text{ km s}^{-1}$: Sobolev $\lesssim 2\%$ vs expansion +38–48%	§5.3
F10	The separation is universal across windows, epochs and ion mixes; realized τ_{max} controls it at fixed v_D and geometry	§5.6
F11	Neglect of light-travel-time evolution is a distinct approximation: frozen $\tau/\tau_S = (1-\beta)/\gamma$ vs worldline $1/\gamma$	App. A7
F12	Overlap is inert in pure absorption (optical depths add exactly); the Sobolev residual is a finite-region boundary effect $\propto v_D/\Delta v_{\text{shell}}$, negligible at thermal widths	App. A6
F13	The expansion-opacity error is bin-width invariant from 0.025 to 41 lines per bin, so it is intrinsic and not a usage artifact	§5.5

mixtures at fixed Doppler width and shell geometry, the separation is universal and the realized τ_{max} is the controlling variable among those axes; the error retains a separate, measured dependence on v_D (§5.4). The error is largest not in fully blanketed bands, where both treatments agree that little escapes, but in the intermediate regime where strong lines have not yet merged—the regime in which realistic calculations still show identifiable spectral features (Domoto et al. 2022; Tanaka et al. 2020). We do not claim this value as the error of a fully mixed kilonova spectrum: F7 shows the error is suppressed as a band goes black, and measuring the mixed case is future work.

The practical implication for kilonova modelling is that “we use the Sobolev approximation” and “we use expansion opacity” should not be treated as interchangeable statements, and that the second deserves a quantified error budget wherever strong lines are present.

Next: worldline-consistent comparison at realistic ejecta velocities (0.1–0.3c), which SEDONA’s frozen steady-state mode cannot supply; and extension to scattering and fluorescence, which is where the analytic equivalence exploited here breaks down and a per-line Sobolev *transport* scheme must be built and tested rather than computed in closed form.

REPRODUCIBILITY

All code, data provenance, executed notebooks, experiment generators, and this manuscript live at https://github.com/zhuygln/revisit_sobolev. Atomic data: GSI Database for Kilonova Radiative Transfer, Zenodo record 19335084 (CC-BY 4.0). Radiative transfer: the public SEDONA release (<https://github.com/dnkasen/pubsed>). Every figure regen-

erates from a committed script; the 68-test suite pins the physics of every module.

APPENDIX A: DERIVATIONS FOR THE NON-SPECIALIST

A1 The formal solution and thermal populations

Multiply Eq. (1) by e^τ with $d\tau = \alpha ds$ and integrate: the intensity leaving a slab is

$$I^{\text{out}} = I^{\text{in}} e^{-\tau_{\text{tot}}} + \int_0^{\tau_{\text{tot}}} S(\tau') e^{-(\tau_{\text{tot}} - \tau')} d\tau', \quad S \equiv j/\alpha. \quad (\text{A1})$$

The first term is the attenuated input beam; the second adds the gas’s own emission, each parcel attenuated by the material still in front of it. The deterministic solver evaluates Eq. (A1) numerically along each ray with $S = B_\nu(T)$, accumulating τ by the trapezoidal rule; convergence requires resolving the Doppler width along the path.

In thermal equilibrium the fraction of ions in level i (energy E_i , statistical weight $g_i = 2J_i + 1$) is

$$\frac{n_i}{n_{\text{ion}}} = \frac{g_i e^{-E_i/k_B T}}{\sum_j g_j e^{-E_j/k_B T}}. \quad (\text{A2})$$

A subtlety our tests caught: the *fraction per statistical weight* is what decreases monotonically with energy; raw fractions need not (at 3000–5000 K a $g = 7$ level at 1016 cm^{-1} out-populates the $g = 5$ ground state of La II).

A2 The Sobolev optical depth

Along a ray toward the observer, the gas at coordinate z moves with line-of-sight velocity z/t (homologous flow), so a photon of observer frequency ν has gas-frame frequency $\nu' = \nu(1 - z/ct)$ to first order in v/c . Insert the line opacity (Eq. 5) into the optical-depth integral (Eq. 2) and change variables from z to ν' ($d\nu' = -(\nu/ct) dz \approx -(\nu_0/ct) dz$ near resonance):

$$\tau = \sigma_{cl} f \int n_l(z) \phi(\nu' - \nu_0) dz \approx \sigma_{cl} f n_l(z_{\text{res}}) \underbrace{\frac{ct}{\nu_0} \int \phi d\nu'}_{=1} = \sigma_{cl} f n_l \lambda_0 t, \quad (\text{A3})$$

which is Eq. (7). The two approximations made are exactly the two Sobolev assumptions: n_l was pulled out of the integral at its resonance-point value (locality), and the profile was integrated as if no other line interfered (isolation).

A3 Why a symmetric gradient cancels (F1)

Let $n_l(v) = n_0[1 + g(v - v_{\text{res}})]$ with g odd about the resonance (e.g. tanh centered there). The resolved integral weights n_l by the even profile ϕ ; the odd term integrates to zero, so the first-order error vanishes. The next term is set by the curvature $n_l''(v_{\text{res}})$ —which for a centered tanh *also* vanishes (inflection point). Only with the resonance off-center does the curvature term survive, giving $E_{\text{Sob}} \simeq \frac{1}{2} |n_l''/n_l| \sigma_v^2 \propto \epsilon^2$, the law measured in Fig. 1.

A4 The expansion-opacity cap (F4)

Integrate the binned opacity of Eq. (8) along the stretch of path over which the moving gas keeps one line inside one frequency bin. The Doppler sweep rate ties path length to bin width, $\Delta z = ct \Delta\nu_{\text{bin}}/\nu$, so the effective optical depth contributed by line k is

$$\tau_k^{\text{eff}} = \alpha^{\text{exp}} \Delta z = \frac{1}{ct} \frac{\nu_k}{\Delta\nu_{\text{bin}}} (1 - e^{-\tau_{S,k}}) \times \frac{ct \Delta\nu_{\text{bin}}}{\nu_k} = 1 - e^{-\tau_{S,k}}. \quad (\text{A4})$$

The bin width cancels: independent of resolution, each resonance crossing contributes at most $\tau^{\text{eff}} = 1$ *per line*, and the transmitted fraction is $e^{-(1-e^{-\tau_S})}$ per line—Eq. (9), the measured 0.421 at $\tau_S = 2$. Note this is *not* a discretization error: it is built into the formalism and no bin refinement removes it. It disappears only when $\tau_S \ll 1$.

A5 Averaging over the source disk (F5’s technical companion)

A resonance plane at height z_{res} in front of a core of radius r_{core} is crossed by the core ray at impact parameter p only if $\sqrt{r_{\text{core}}^2 - p^2} < z_{\text{res}} < \sqrt{r_{\text{out}}^2 - p^2}$. The observed attenuation is therefore

$$\frac{F}{F_{\text{cont}}}(\nu) = \frac{2}{r_{\text{core}}^2} \int_0^{r_{\text{core}}} p \exp\left[-\sum_{k \text{ crossed}(p,\nu)} \tau_k\right] dp, \quad (\text{A5})$$

not $e^{-\sum \tau}$ with a global crossing criterion: planes at $z_{\text{res}} < r_{\text{core}}$ shadow only part of the disk. Ignoring this ($\sim$ plane

counting) predicts measurably too-shallow troughs in the ladder experiment.

A6 Boundaries, not overlap: where the Sobolev error comes from

Two Sobolev-intrinsic assumptions could in principle fail: locality and isolation. Neither produces the residual measured in §5.3.

A6.0.1 Isolation cannot fail here. With fixed populations and pure absorption the opacities of different lines simply add, so

$$\tau_\nu = \int \sum_i \alpha_{\nu,i} ds = \sum_i \int \alpha_{\nu,i} ds = \sum_i \tau_{\nu,i}, \quad (\text{A6})$$

and the emergent intensity is $I_{\nu,0} \exp(-\sum_i \tau_{S,i})$ whether the profiles are separated or lie exactly on top of one another. Driving two identical lines from 20 Doppler widths apart to exact coincidence reproduces the Sobolev result to six decimal places at every separation. Overlap becomes a real effect only when scattering, fluorescence or NLTE populations allow one transition to change the radiation field another one samples; those are switched off here by construction.

A6.0.2 What does fail is geometric. The line-forming region is finite. A resonance at z_{res} in a shell spanning z_{lo} to z_{hi} has its profile clipped by the edges, so for a central ray

$$\frac{\tau_{\text{resolved}}}{\tau_S} = \frac{1}{2} \left[\text{erf}\left(\frac{z_{\text{hi}} - z_{\text{res}}}{\ell_S}\right) - \text{erf}\left(\frac{z_{\text{lo}} - z_{\text{res}}}{\ell_S}\right) \right], \quad (\text{A7})$$

with the Sobolev length $\ell_S = v_D t$. This is unity deep inside, exactly 1/2 at an edge and zero outside—while the Sobolev prescription applies a step of 1 inside and 0 outside. Direct integration reproduces this to four decimals at every width tested (Fig. A1, left), and the curves collapse when plotted against d/v_D , confirming that the width enters only through the ratio.

The band-averaged consequence follows: the affected fraction of the spectrum is the fraction of the velocity span lying within a few Doppler widths of an edge. With two edges that is $\sim 2v_D/\Delta v_{\text{shell}}$, so for our shell ($\Delta v_{\text{shell}} = 2000 \text{ km s}^{-1}$) it runs from 1% at $v_D = 10 \text{ km s}^{-1}$ to 30% at 300 km s^{-1} . This is an upper bound on the band error rather than a prediction of it, since only the clipped part of each profile is misassigned: the measured $|\Delta_{\text{Sob}}|$ over the same range is 0.2% and 9.2% (Fig. A1, right). What the estimate does capture, and what matters here, is the linear scaling in $v_D/\Delta v_{\text{shell}}$ and hence the two orders of magnitude by which the effect shrinks at physical line widths.

This is worth stating plainly because it is easy to mistake for physics: the error is a property of imposing a *finite* line-forming region on a calculation whose approximation assumes a locally infinite one. In a real ejecta the same effect appears at the photosphere and at the outer edge of the line-forming zone, and it scales with the thermal width—which for lanthanides is $\sim 0.6 \text{ km s}^{-1}$ against ejecta spans of 10^4 km s^{-1} , i.e. utterly negligible. Our sensitivity to it at $v_D = 100\text{--}300 \text{ km s}^{-1}$ is an artifact of using artificially broad lines to keep the frequency grids affordable.

[t]

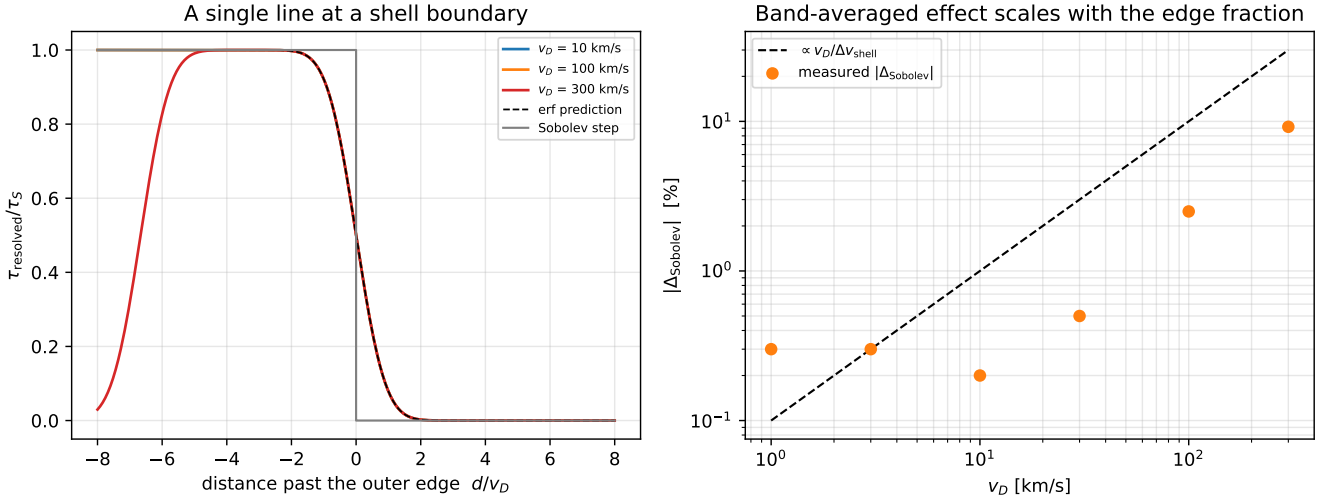


Figure A1. Left: a single line’s resolved optical depth as its resonance is walked out through the outer edge of the line-forming region, in units of the Doppler width. The curves for $v_D = 10, 100$ and 300 km s^{-1} coincide — the effect is self-similar in d/v_D — and follow the error-function prediction exactly, against the hard step that Sobolev applies. The 300 km s^{-1} curve departs from unity on the left because the shell is then only ~ 7 Doppler widths wide and the inner edge clips as well. Right: the band-averaged consequence tracks $v_D/\Delta v_{\text{shell}}$.

A7 Frozen snapshot versus photon worldline: what F3 really was

This appendix was wrong in the previous draft, in an instructive way. We keep the faulty step visible because it is easy to make and hard to see.

Appendix A2 derived τ_S to first order in v/c . Kilonova ejecta move at $0.1\text{--}0.3c$, and we measured (Finding F3) that a resonance at $z_{\text{res}}/ct = 7.7\%$ gave a trough matching $\exp[-\tau_S(1 - z_{\text{res}}/ct)]$ to four digits rather than $\exp(-\tau_S)$. The natural conclusion—which we drew, and which an earlier draft of this appendix argued—was that this is the leading relativistic correction and that it is $\mathcal{O}(v/c)$. It is not.

A7.0.1 The faulty step. That derivation differentiated the Doppler factor $D = \gamma(1 - b)$ at fixed t , i.e. it pushed a photon through a frozen snapshot of the ejecta. But a photon takes time to cross a resonance, and in homologous flow $\beta = \mathbf{r}/(ct)$ depends on the age as well as the position. Along the photon worldline $\mathbf{r}(t) = \mathbf{r}_0 + c\hat{\mathbf{n}}(t - t_0)$,

$$\frac{d\beta}{dt} = \frac{\hat{\mathbf{n}} - \beta}{t} \implies \frac{db}{dt} = \frac{1 - b}{t}, \quad (\text{A8})$$

whereas the frozen calculation uses $db/dz = 1/(ct)$. The two differ by a factor $(1 - b)$ —first order in β , the same order as the effect being computed. The time term does not vanish because the crossing is brief: what matters is the *gradient*, and the time term is proportional to b itself.

A7.0.2 The correct law. With $D = \gamma(1 - b)$ one finds $dD/dt = -D\gamma^2(1 - b)/t$. Using the opacity transformation $\chi_{\text{lab}} = D\chi'$ (from the invariance of $\nu\chi_\nu$) and the resonance condition $D\nu = \nu_0$, the profile integral collapses to $\tau = D\chi'c/|d\nu'/dt|$, and the factors of $\gamma(1 - b)$ cancel down

β	frozen τ/τ_S	worldline τ/τ_S	$1/\gamma$
0.05	0.94881	0.99875	0.99875
0.10	0.89549	0.99499	0.99499
0.20	0.78384	0.97980	0.97980
0.30	0.66776	0.95394	0.95394

to a single Lorentz factor:

$$\frac{\tau_{\text{rel}}}{\tau_S} = \frac{1}{\gamma} = \sqrt{1 - \beta^2} \quad (\text{A9})$$

with τ_S evaluated using the density and ejecta age local to the resonance, which makes the statement independent of how the calculation is anchored in time. **There is no first-order term:** $1/\gamma = 1 - \frac{1}{2}\beta^2 + \mathcal{O}(\beta^4)$. The frozen calculation instead gives, for radial rays, $\tau_{\text{frozen}}/\tau_S = (1 - \beta)/\gamma$.

A7.0.3 Numerical confirmation. Integrating the resolved line opacity directly along the path—no Sobolev assumption anywhere—reproduces both expressions to five decimal places, with the ejecta age held fixed or allowed to advance:

So the relativistic correction at $\beta = 0.1, 0.2, 0.3$ is 0.5%, 2.0%, 4.6%—not the 10%, 22%, 33% that the frozen law implies and that the previous draft of this appendix claimed.

A7.0.4 Which problem does the Monte Carlo code solve? It is natural to assume a Monte Carlo transport code supplies the time-dependent answer, and we briefly did. It does not, in the configuration used here. Sweeping a single line’s resonance velocity from $\beta = 0.001$ to 0.34 and comparing the measured $\tau_{\text{eff}} = -\ln(F/F_{\text{cont}})$ against both laws gives an RMS deviation over $\beta_{\text{res}} \gtrsim 0.1$ of 0.024 from the frozen law and 0.55 from the worldline law—a factor 23—and at $\beta_{\text{res}} = 0.34$ the measurement is 0.613 against a frozen prediction of 0.618 and a worldline prediction of 1.43 . SEDONA run

with `transport_steady_iterate` iterates the radiation field on a fixed grid at fixed epoch; its source disables hydrodynamics in that mode and reads no time-stepping parameters. It is a frozen-snapshot calculation.

This matters for the harness. Cross-code comparison requires matching the *transport treatment* as well as the thermal-emission convention (F8) and the spectral normalization (§5.6): a third convention, and the one with the largest lever arm at high velocity. Like-for-like against a frozen SEDONA means running the solver in its frozen mode. At the shell velocities used for the measurements in this paper ($\beta \leq 0.01$) the two laws differ by $\lesssim 2\%$ in τ . That is far below the level at which any Δ_{exp} moves—those are same-code differentials—but it is comparable to the $\sim 1\%$ resolved-leg agreement itself, so every solver-versus-SEDONA comparison in this paper is quoted with the solver in frozen mode, matching SEDONAS transport treatment. At kilonova velocities the choice would dominate.

A7.0.5 What F3 is, finally. Not a frame ambiguity (our first reading), not the leading relativistic correction (our second reading), but an artifact of integrating a frozen snapshot—a problem the photon does not solve. The distinction earns its own entry in Table 2 as F11, because “neglect of light-travel-time evolution” is a separate approximation from the Sobolev localization, from the independent-line assumption, and from expansion-opacity binning. Its practical consequence for the measurements reported here is that at our shell velocities the correct correction is 5×10^{-5} : negligible, and therefore *not* an explanation for the Sobolev residual of §5.3.

For the general case the resonance locus is not a plane— γ depends on r , not z —and one needs the common-direction and common-point frequency surfaces of Jeffery (1993). Our controlled experiments keep the emitting core small compared with the resonance radii, so all core rays are nearly radial and the plane picture holds.

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